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UNIT 1:

Semi conductors diodes and application

Electronics :- It is a branch of Engineering which deals with the study of flow of electrons.

* Electrons was invented by J.J. Thomson (1897).
* Conductors are the materials or substances which allows electricity to flow through them.
* Conductors will be having large numbers of free electrons
* Conductors exhibit highest conductivities.
  Ex: Cu, Al etc...
* Insulators are the materials or substances which doesnot allows electricity to flow through them.
* Insulators will not be having large numbers of free electrons.
* Insulators exhibit lowest conductivities.
  Ex: Rubber, Plastic, Wood etc...,
* Semi Conductors are the materials whose conductivities lies between conductors and insulators.
  Ex: Silicon ; Germanium.
* Conductivity in semiconductor is due to the presence of valence electrons present in outermost shell or orbit.

Types of semiconductors : These are the pure semiconductors without any impurity.

1. Intrinsic Semiconductors - These are the pure semi conductors without any impurity.
2. Extrinsic semiconductors - These are the semiconductors which requires the impurities to be added in order to increase its conductivities.

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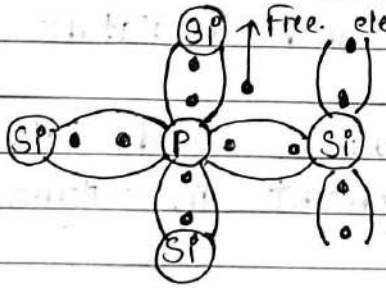
The process of adding impurity to semiconductor is called doping.
The impurity added is called dopant.

There are two types of extrinsic semiconductors
N type
P type
which are based classified based on -ties added.

N type semiconductors : These are created by doping intrinsic semiconductors with the donor impurities pentavalent impurities.

* Common dopant is phosphorous.
* In n-type there will be large electron concen and less hole-concentration.
* N-type refers to the negative charge carriers i.e., electrons.
* In N-type es are the majority charge carr and holes are the minority charge carriers.

P-5 valence electrons, Si-4

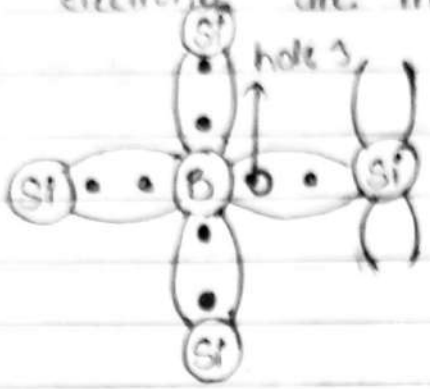


P type semiconductors : These are created by doping Intrinsic semiconductors with trivalent donor impuri

* Common dopant used is Boron.
* Here, the holes concentration will be large tha electron concentration.
* P-type refers to positive charge carriers i.e holes.

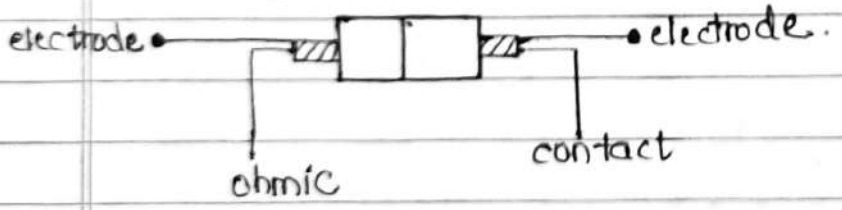
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* In p type holes are the majority charge carriers and electrons are minority charge carriers.

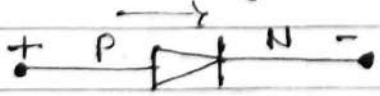


P-N Junction diode.

* Two type of material namely p-type and n-type are chemically combined with a special fabrication technique to form a p-n junction.
* P-N junction provided with a copper wire connecting leads to becomes an electronic device known as p-n junction diode.



Symbol of p-n junction diode:



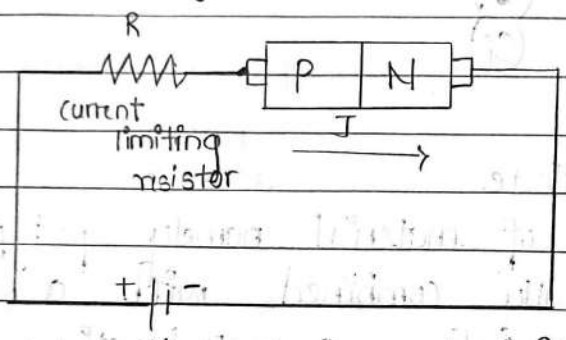
* Diode is a two terminal, semiconductor device which allows the current to flow in only one direction.
* Hence, it is called as one-way device.
* It offers low resistance when it is forward bias and high resistance when it is reverse bias.
* Anode (+), Cathode (-) are two terminals.

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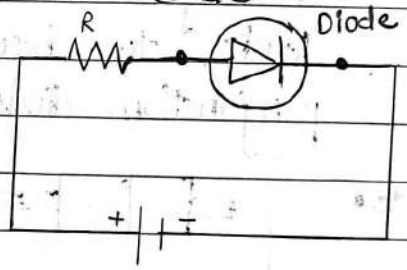
Operation of a Diode :

The process in which an external DC voltage or a battery is connected to the P-N junction diode is called as biasing.

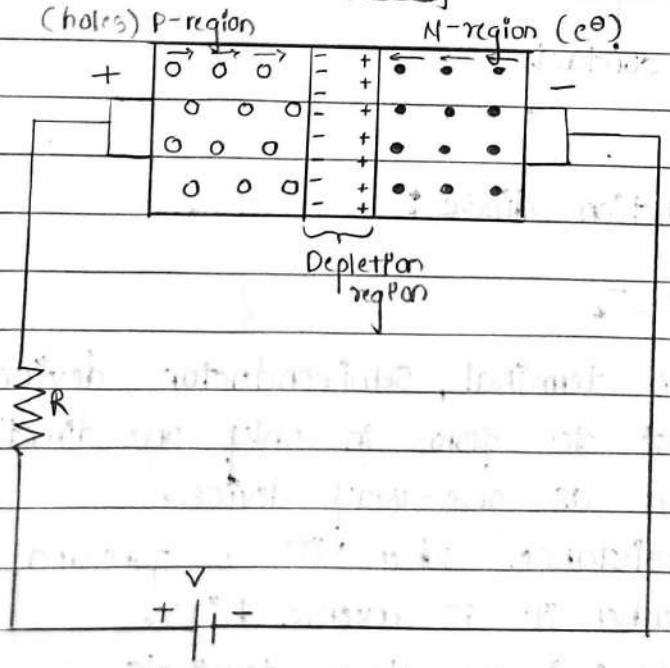
forward biasing diode :

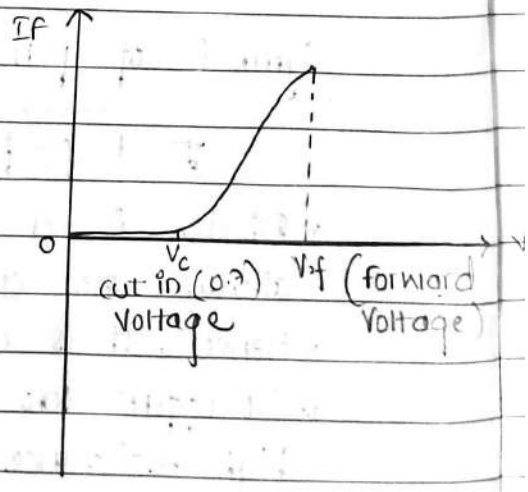


Symbolic representation of diode



forward biasing





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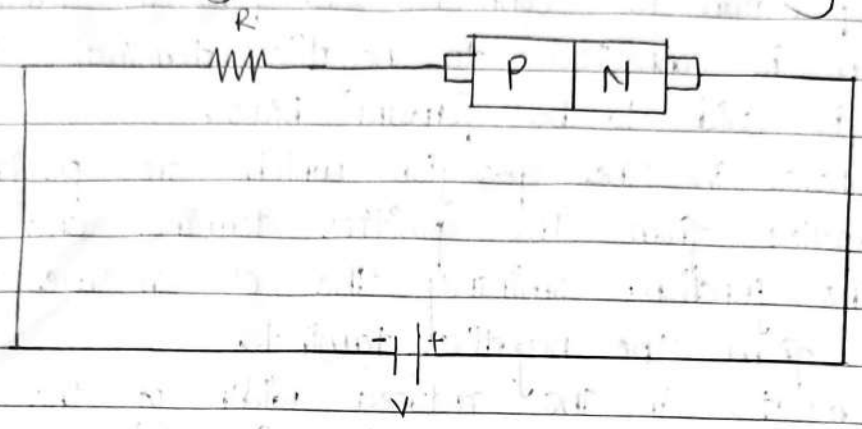
* If an external voltage is connected in such a way that p-region is connected to positive terminal & n-region is connected to negative terminal then the diode is said to be forward bias.
* The Holes in the p-region which are positively charged are repelled from the positive terminal and move towards the junction. Similarly the $e^{\ominus}$ in the n-region will repelled from the negative terminal.
* This results in the reduced width of the depletion region and barrier potential is also reduced.
* If the applied voltage is increased from 0 the width of the depletion region is further reduced and very small current flows through diode.
* When the forward voltage (V_F) is greater than the cut in voltage (Knee voltage) (threshould voltage) the depletion region gets completely collapsed and large number of charge carriers crosses the junction.

So, the current increases drastically and that current is called forward current.

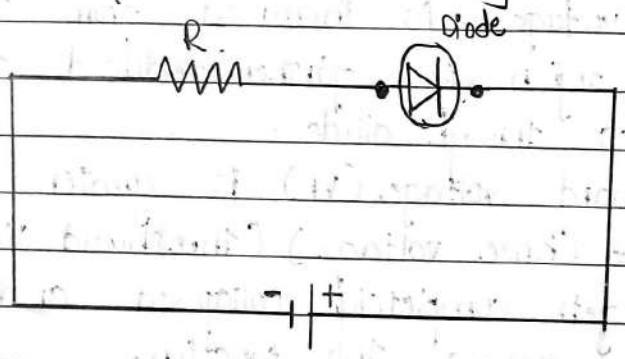
* This forward current is due to majority charge carriers and a current limiting resistor is used to limit highly increasing forward current.
* ~~This forward current is due to majority charge carriers and a current limiting resistor is used to limit highly increasing forward current.~~
* Silicon cut in voltage - 0.7V, ~~Uranium~~ Germanium - 0.3V
Forward Static resistance, $R_f = \frac{V_f}{I_f}$ Ω

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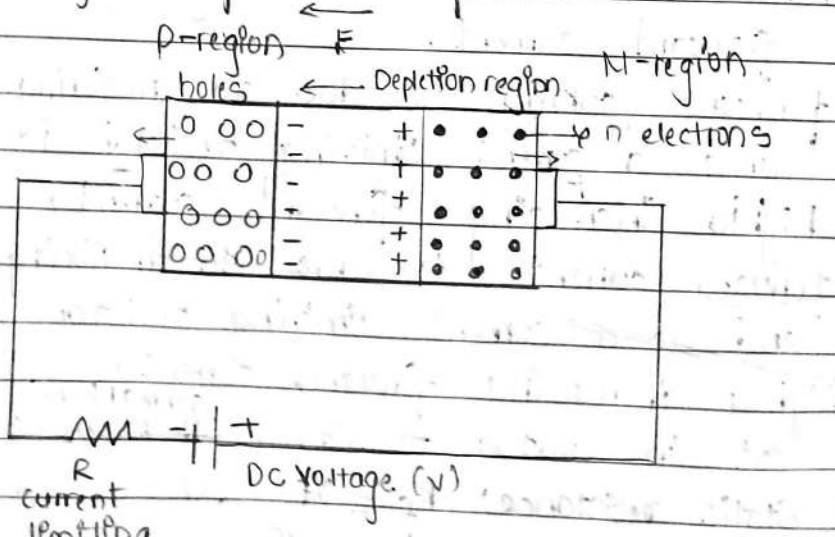
Reverse biasing [operation of a diode during reverse bias



Reverse biasing



Symbolic representation of diode



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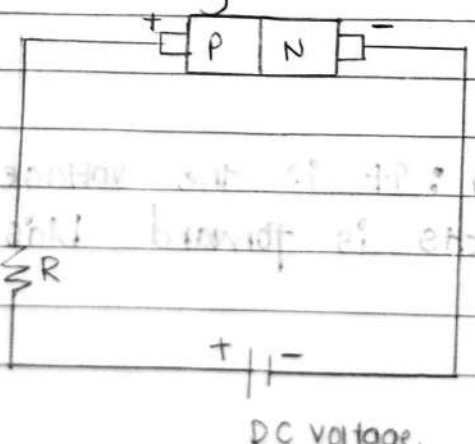
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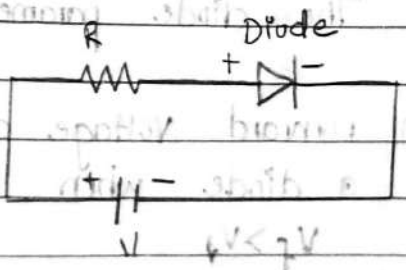
* If an external dc voltage is connected in such a way that n-region is connected to the positive and p-region is connected to the negative terminal, then the diode is said to be reverse biased.
* The holes in the p-region which are positively charged are attracted from the negative terminal and move away from the function. similarly, the electrons in the n-region which are negatively charged are attracted from the positive terminal and move away from the function.
* Hence, the width of the depletion region increase with the increase in the reverse voltage.
* If the reverse voltage increases, the width of the depletion region increases with and a very small current flows due to the minority charge carriers. This reverse current is called reverse saturation current.
* If the reverse voltage increases beyond a particular value, large reverse current flows and the diode goes into reverse break down. This may damage the diode. Hence, reverse breakdown is avoided in practice.

Reverse static resistance, $R_r = \frac{V_r}{I_r}$ Ω.

V-I characteristics of a diode:

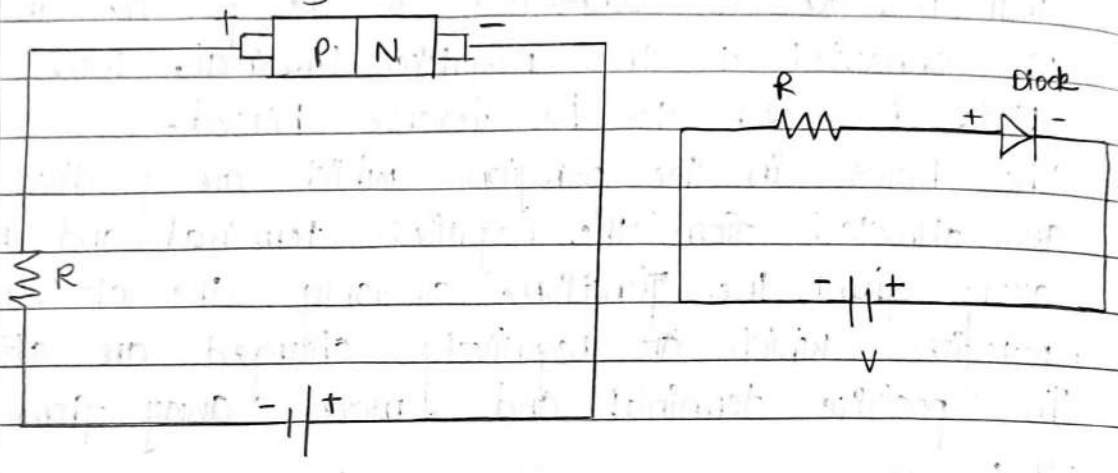
Forward biasing

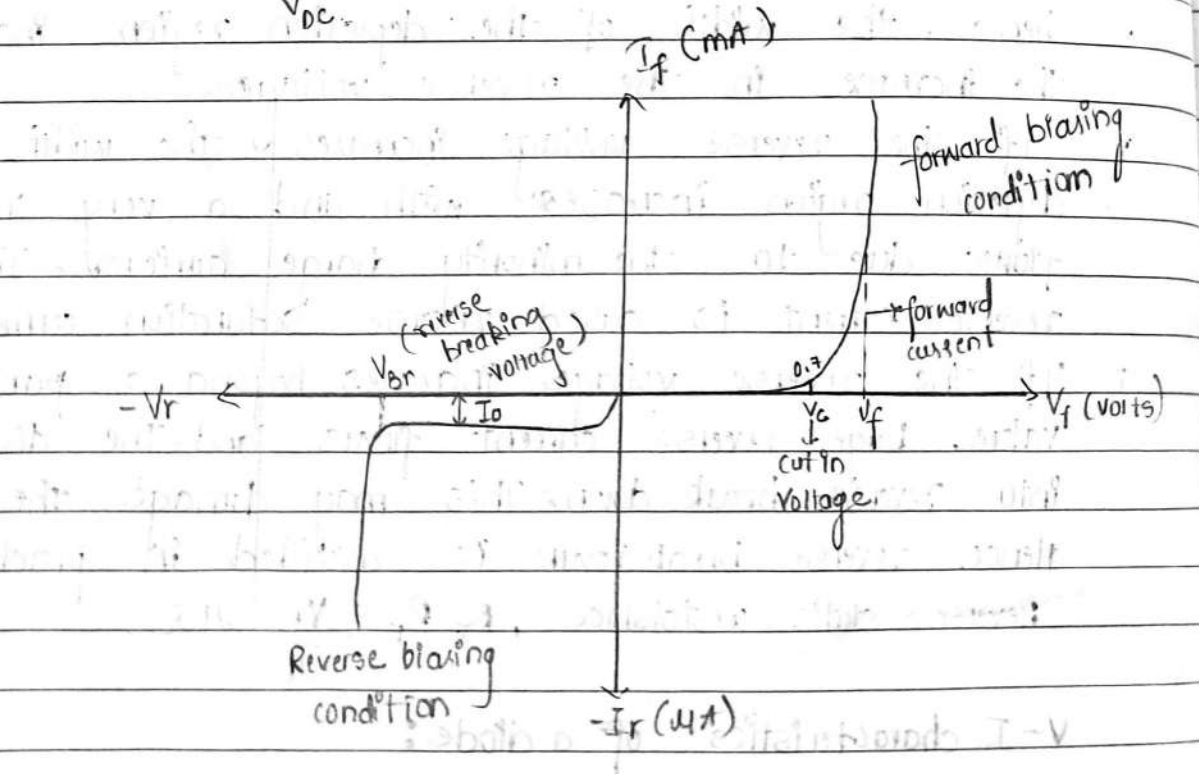




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Reverse biasing





Diode parameters:

The diode parameters are

i) forward voltage drop (V_f) : It is the voltage drop across a diode when it conducts is forward bias voltage(V_f).

$V_f > V_\gamma$

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for silicon diode, $V_\gamma = 0.7V$ and
for Germanium diode $V_\gamma = 0.3V$

ii) Maximum forward current (I_f or I_{fmax}): It is the maximum current the diode can pass under a forward bias condition, without damage to p-n junction due to overheating.

iii) Reverse saturation current (I_R) or (I_o):
It is the current which flows through the diode when it is reverse biased.
for silicon diode, I_R is in the order of nA.
For germanium diode, I_R is in the order of μA.

iv) Reverse break down voltage (V_{BR}):
It is the reverse bias voltage at which the p-n junction break down permanently and damage the diode.
for silicon diode $V_{BR} < 50V$
for Germanium diode V_{BR} is about 100V.

v) Reverse recovery time [t_{rr}]:
It is defined as the time taken by the diode in reverse bias condition to turn off from on state.

vi) Maximum power dissipation (P_{Dmax}):
The product of voltage drop across the diode & current flows through the diode is the power dissipation across the diode.
$P_{Dmax} = V_D \times I_D$.

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Diode current equation:

$$I = I_o \left[e^{\frac{V}{nV_T}} - 1 \right]$$

where,

I = current flowing through the diode.

I_o = Reverse saturation current.

V = applied voltage

V_T = temperature dependent voltage.

n = constant

Si = 2 ⇒ Ge = 1

$V_T = KT$ $\quad V_T = \frac{KT}{q}$

K = Boltz man constant $= 1.38 \times 10^{-23}$

T = Temperature in kelvin

q = charge $= 1.6 \times 10^{-19} C$.

Hence,

$$I = I_o \left[e^{\frac{qV}{nKT}} - 1 \right] A.$$

Effects of temperature on a diode:

The temperature has the following effects on the diode parameters.

* Cut-in voltage decreases as the temperature increases, Hence, the diode conducts at smaller voltage at larger temperature.
* Rerverse saturation current increases as the temperature increases.

Normally, reverse current I_o ~~dou~~ doubles at every 10°C raise in temperature.

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$I_{O_2} = 2^{(\Delta T/10)} \cdot I_{O_1}$

where, I_{O_2} = Reverse current at $T_2°C$

I_{O_1} = Reverse current at $T_1°C$

$\Delta T = (T_2 - T_1)$

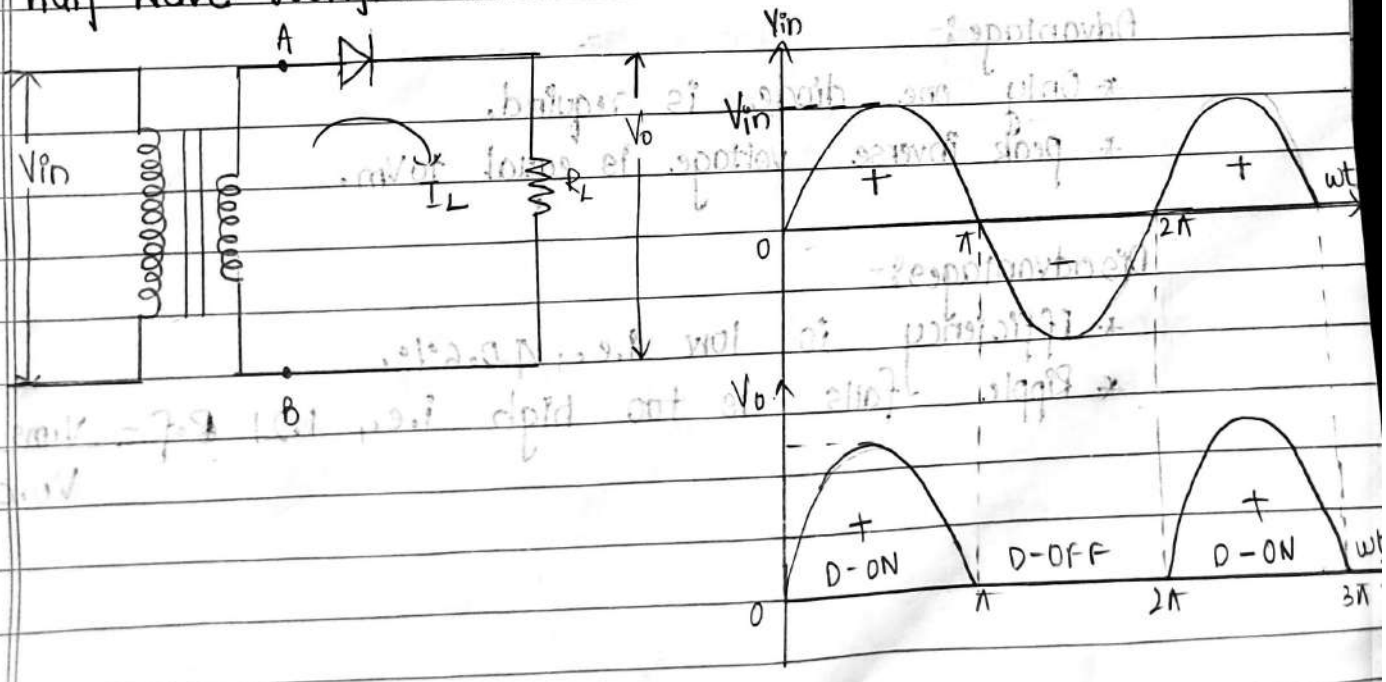
* Reverse breakdown voltage increases as temperature increases.
* As the temperature increases the voltage equivalent of temperature also increases. Hence, dynamic forward resistance increases.

Voltage equivalent of temperature (V_T) indicates dependence of diode current on temperature.

Rectification :-

* Rectification is the process of converting alternating current into the direct current.
* Rectifiers is a device which converts alternating current in to the pulsating direct current.

Half wave rectifier (HWR):-



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* Half wave rectifier consists of a single diode in series with the load resistors as shown in the figure.
* When AC input is applied to the primary winding of the transformer, It is coupled across Secondary.
* During +ve half cycle of the AC input end A is +ve with respect to end B. Diode D is forward biased and act as a short circuit. Hence the current flows through R_L and output follows the input.
* During -ve half cycle of the AC input end A is -ve with respect to end B diode D is ~~forward~~ reverse biased and act as a ~~short~~ open circuit. Hence, ~~the current flows through~~ R_L and output follows the input. No current flows through R_L. $\therefore V_o = 0$.

The DC input wave form is expected to be straight line but the output is in the form of +'ve sinusoidal pulse.

Hence, the output is pulsating DC.

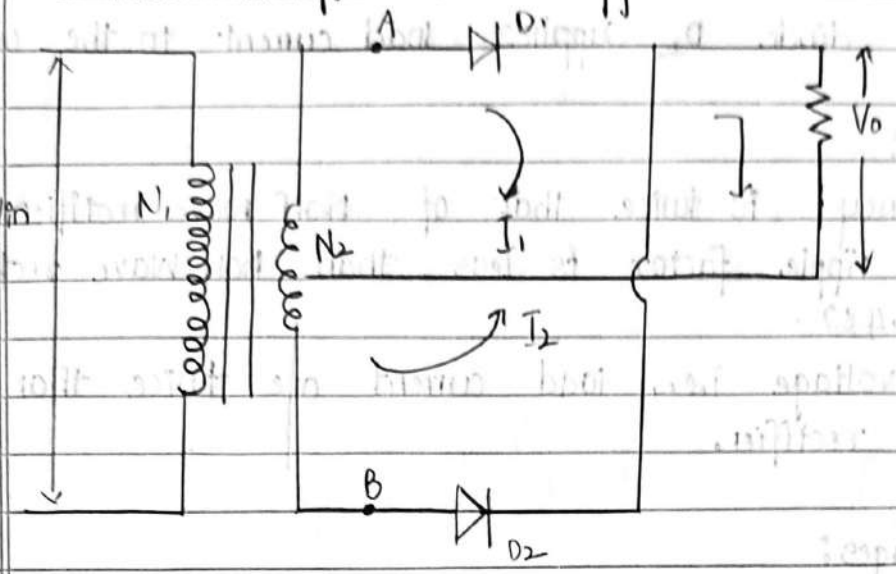
Advantage:-
* Only one diode is required.
* peak Inverse voltage is equal to V_m.

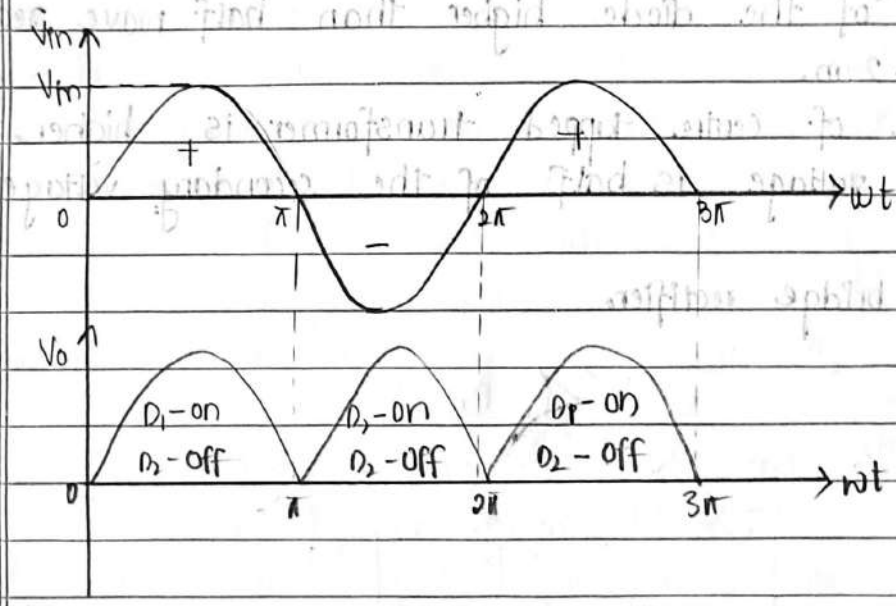
Disadvantages:-
* Efficiency is low i.e., 40.6%.
* Ripple falls is too high i.e., 1.21 $R_f = \frac{V_{rms}}{V_{D.C}}$

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full Wave rectifiers (centre tapped) :





* full wave rectifier consists of two diodes D_1 and D_2 connected across centre tapped secondary and load resis-tance as shown in the fig.
* During +ve half cycle of AC input voltage end A is +ve wrt end B is forward biased and conducts, while diode D_2 is reverse biased and will not conduct thus Diode D_1 supplies load current to the output.
* During -ve Half cycle of AC input voltage end A is -ve wrt to end B. Since, D_2 is forward biased & conducts,

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while diode D_1 is reverse biased & will not conduct thus the diode D_2 supplies load current to the output.

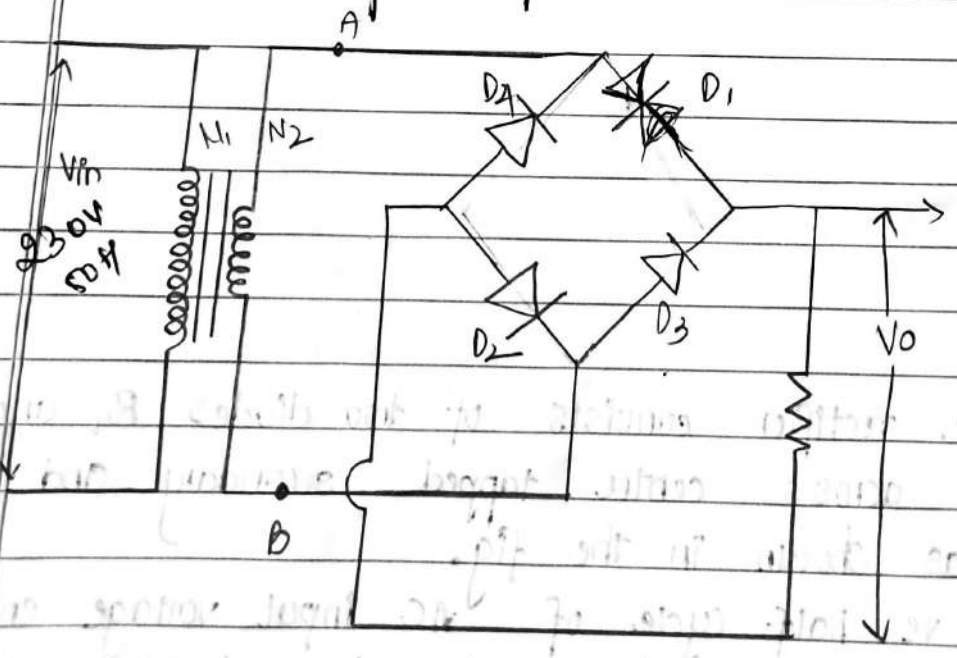
Advantages:

* Efficiency is twice that of Half wave rectifier 81.2%
* The ripple factor is less than half wave rectifier is 0.482.
* DC voltage i.e., load current are twice than Half wave rectifier.

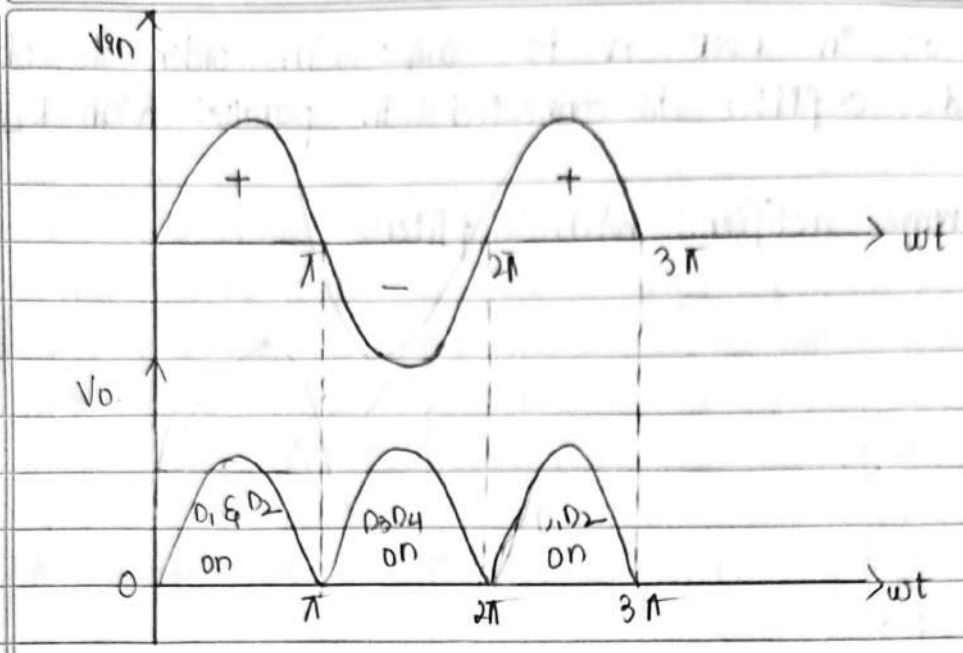
Disadvantages:

* PIV of the diode higher than half wave rectifier PIV = 2Vm.
* Core of centre tapped transformer is higher.
* O/P voltage is half of the secondary voltage.

Full Wave bridge rectifier.



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* Bridge rectifiers consists of four diodes D_1, D_2, D_3, D_4 connected across secondary transformer and load resistance these four diodes forms a bridge network as shown in the figure when AC input is applied to the primary winding of the transformer. It is coupled across secondary
* During '+ve' half cycle of the AC input voltage, end A is '+'ve with respect to end B. This makes the diodes D_1 & D_2 forward biased and conducts while the diodes D_3 & D_4 are reverse biased.
* During '-'ve half cycle of the AC input voltage, end A becomes '-'ve with respect to end B. This makes diode D_3 D_4 forward biased & conducts while the diodes D_1 & D_2 are reverse biased. The output wave form is shown in the figure.

Rectifiers with C-filters.

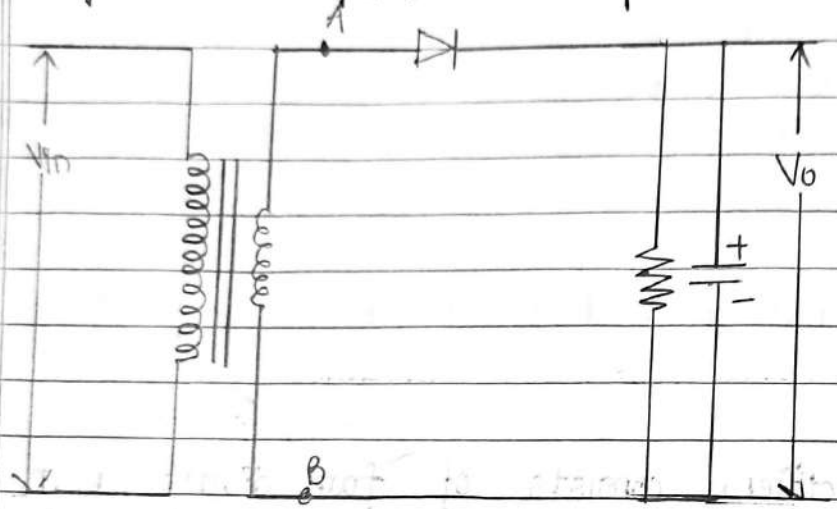
filters is a circuit used to reduce the ripple content present in the rectified output.

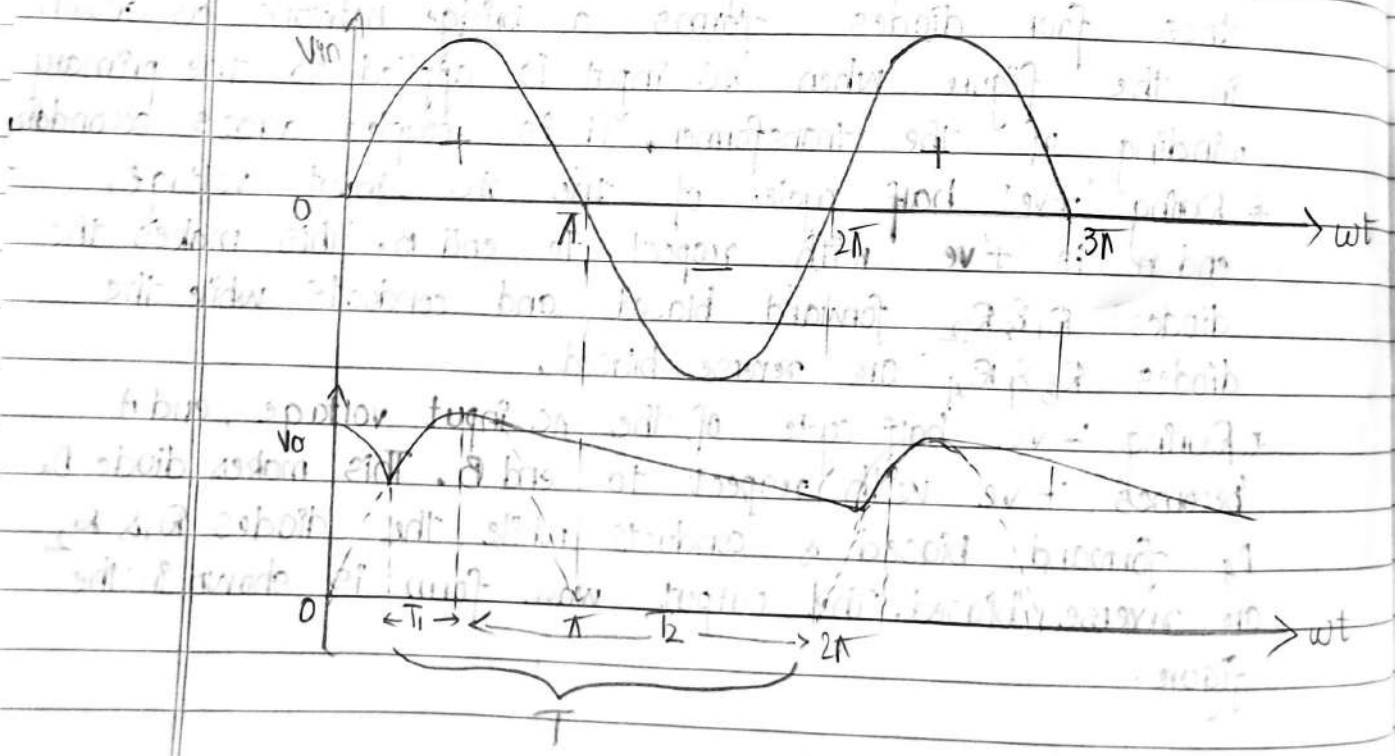
The output from the rectifiers is not pure dc due to ripple content. In HWR ~~rectifi~~ ripple content is 1.21

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where as in FWR it is 0.48. In order to obtain pure dc, c-filter is connected in parallel with R_L.

Half wave rectifiers with 'c' filters





During +'ve half cycle of the A.C input, end A is +ve with respect to 'B'. Hence, the diode D is forward biased and conducts. Thus the capacitor charge to the peak value of V_m of the input voltage.

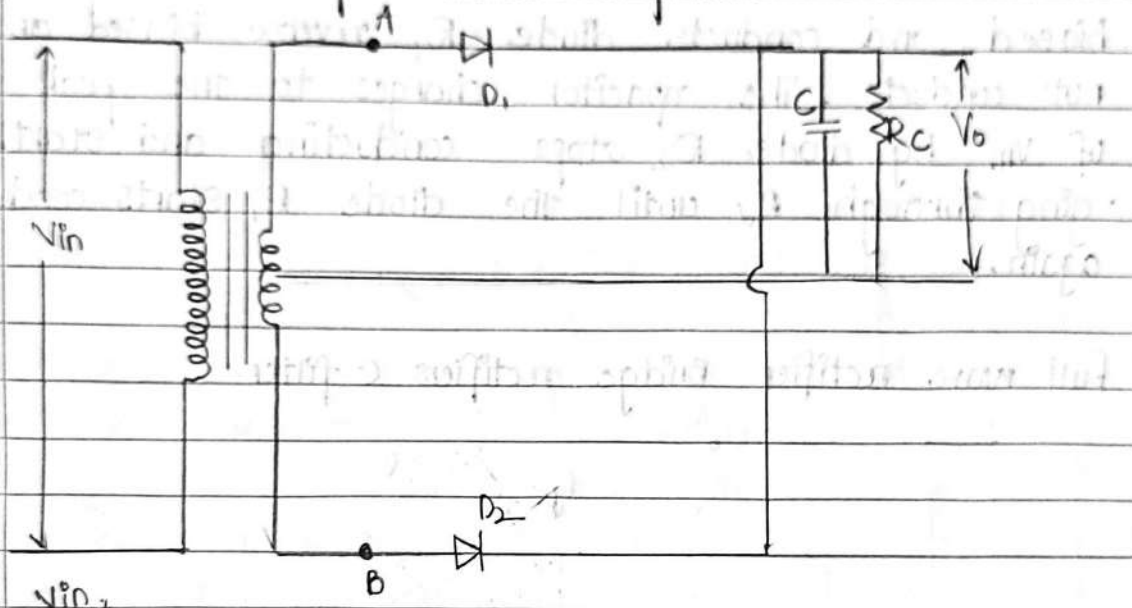
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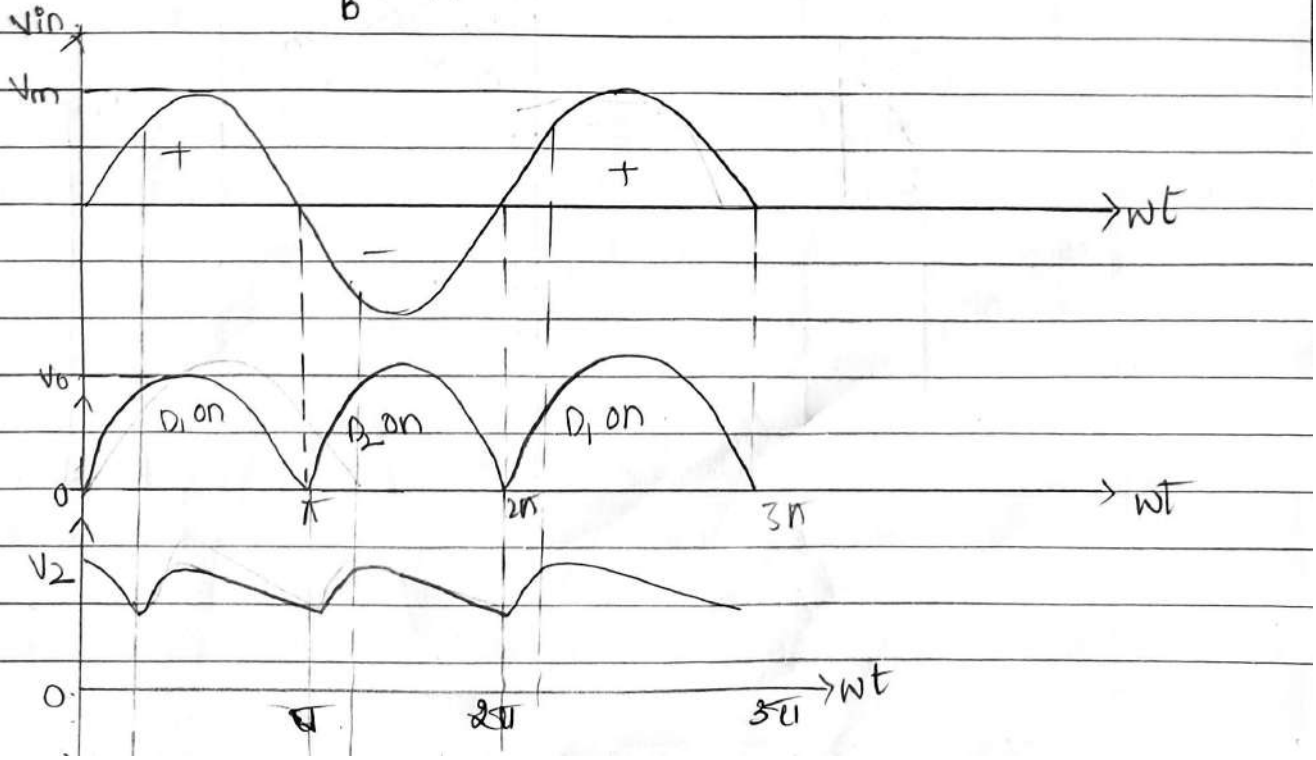
when the input voltage falls below V_m, the diode stops conducting and the capacitor starts discharging through R_L.

The discharging of capacitor continues till the diode starts conducting again.

During -'ve half cycle of the AC input, end A is negative w.r.t B. Hence, the diode D is reverse biased and it doesnot conducts.

Full wave rectifier with 'C' filter.



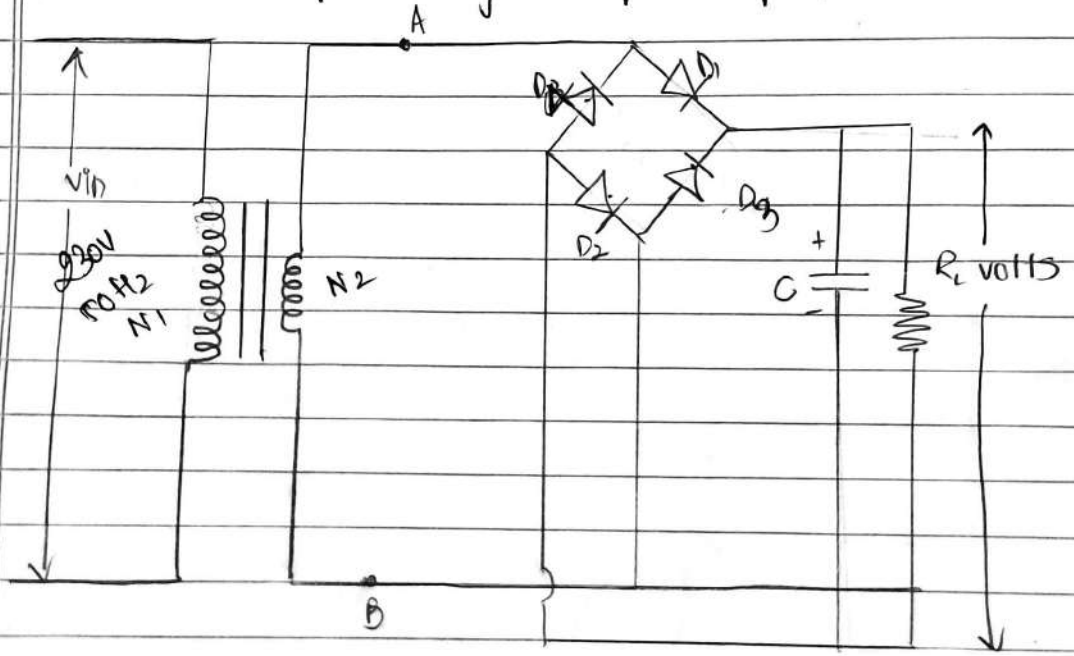


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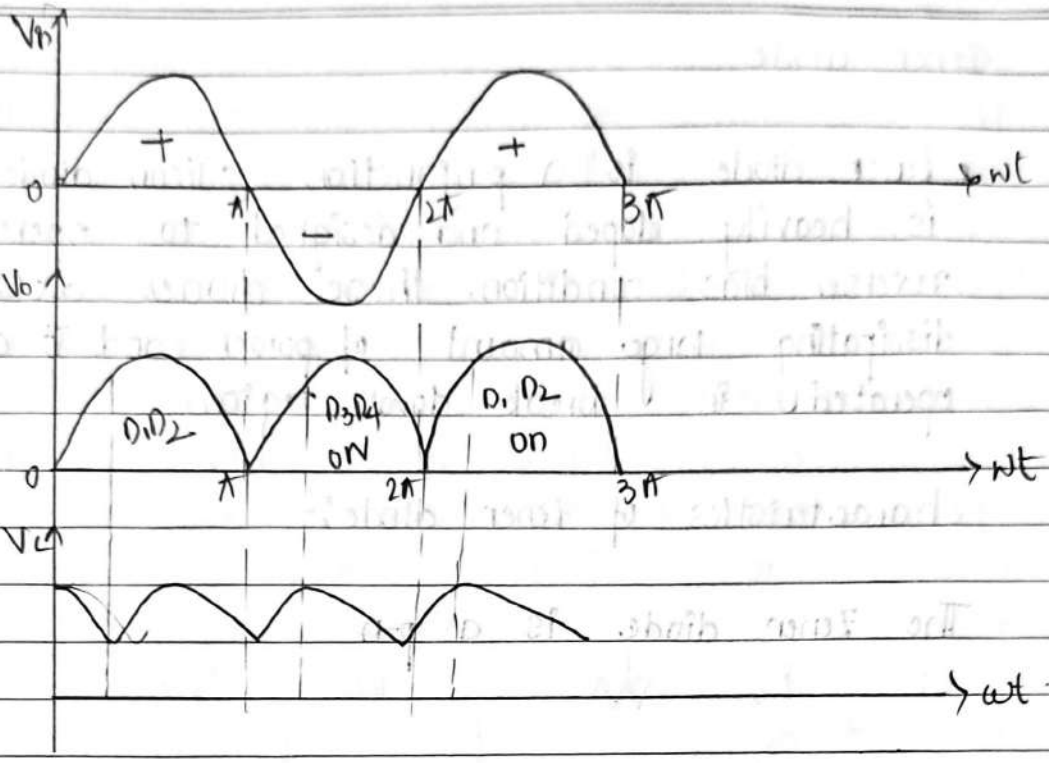
* During positive half cycle of the AC input, end A is +ve wrt to end B. hence the diode 'D' is forward and conducts. Thus, the capacitor charges to the peak value of V_m of the input voltage, when the input voltage falls below V_m, the diode D_1, stops conducting and starts discharging through R_L. The discharging capacitor continues until diode D_2 stats conducting next half cycle.

* During -'ve half cycle of the AC input end 'B' becomes +'ve wrt end A. Hence, the diode D_2 forward biased and conducts diode D_1 reverse biased and will not conducts. The capacitor charges to the peak value to of V_m by diode D_2 stops conducting and starts discharging through R_L until the diode D_1 starts conducting again.

Full wave rectifier Bridge rectifiers C-filter.



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During +'ve half cycle of the AC input, end A becomes positive wrt to end 'B', the diode D_1 & D_2 are forward biased and D_3 & D_4 are reverse biased. Hence D_1 & D_2 conducts and charges the capacitor to the peak value of V_m. When the input voltage falls below V_m, the diodes stops conducting and capacitor starts discharging through R_L until diodes D_3 & D_4 starts conducting again in the next half cycle of the input.

During -'ve half cycle of the AC input, end B is +ve with respect to end A. The diodes D_3 & D_4 are formed biased. D_1 & D_2 reverse biased. Hence, diosedes D_3 & D_4 conducts and charges the capacitor to the peak value of V_m. When the input voltage falls below V_m the capacitor starts discharging through R_L until diodes D_1 & D_2 starts conducting again in the next half cycle.

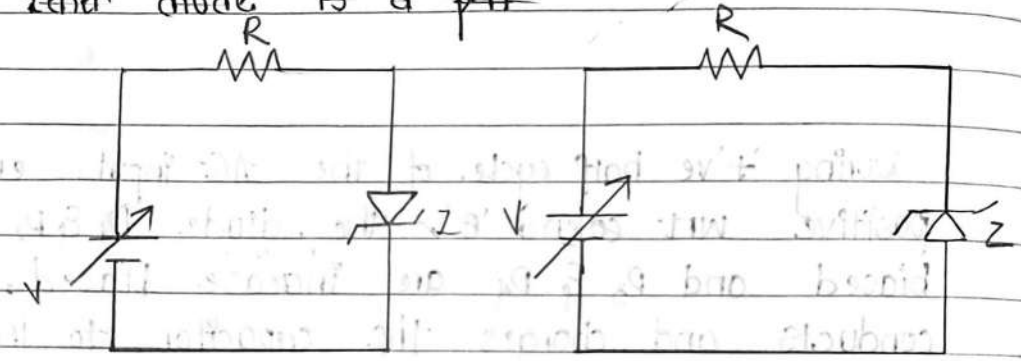
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Zener diode

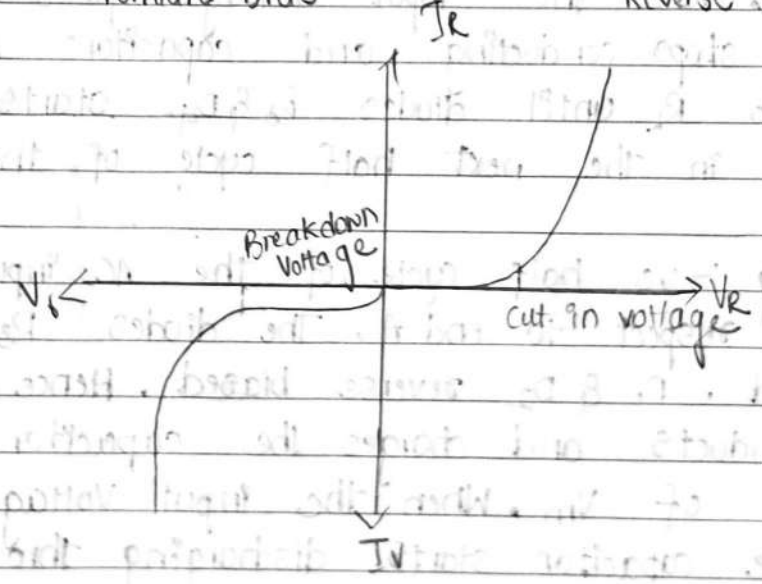
* Zener diode is a p-n junction silicon diode which is heavily doped and designed to operate under reverse bias condition. These diodes are capable of dissipating large amount of power and it can be operated in break down region.

characteristics of Zener diode:-

The Zener diode is a ~~p-n~~



forward bias Reverse bias



* In the forward bias condition, the characteristics of Zener diode is similar to ordinary diode but the zener diode is designed to operate in reverse

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biased condition.

* In reverse biased condition when reverse voltage is applied to zener diode is increased. Diode carries very small current called reverse saturation current till the reverse voltage applied is less than the reverse breakdown voltage.

* When the reverse voltage ~~exists~~ exceeds reverse breakdown voltage the current through it changes drastically but the voltage across it remains constant.

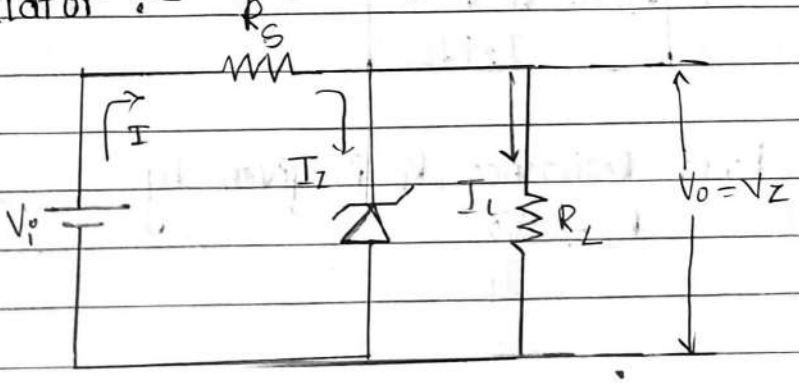
Such a breakdown region is the normal operating region for a zener diode.

* The maximum current a zener diode can carry safely is called zener maximum current and it is denoted by I_{ZM} [maximum zener current].

* The minimum current flows through zener diode upto ~~greater~~ breakdown voltage is called zener minimum current (I_{ZK}).

* Zener diode has voltage regulator [stabilizer].

Voltage regulator :-



Zener diode is used as a voltage regulator because it maintains the constant output voltage even though the current through it changes.

Zener diode is used in reverse bias region under this condition the current through the diode is very small when sufficient reverse voltage is applied zener breakdown occurs & the large current flows

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through the diode.

The voltage at which zener diode conducts is called zener voltage (V_z). Under this condition the voltage across the zener is constant and equal to V_z. Therefore, $V_o = V_z$.

From the circuit, $I = I_z + I_L$ (KCL)

$$\boxed{I_z = I - I_L} \rightarrow (1)$$

The current I, flows through the resistor R is given by

$V_i = R_s I + V_o$

$R_s I = V_i - V_o$

$$\boxed{I = \frac{V_i - V_o}{R_s}} \rightarrow (2)$$

substitute (2) in (1)

$$I_z = \frac{V_i - V_o}{R_s} - I_L$$

$$I_z + I_L = \frac{V_i - V_o}{R_s}$$

$$\boxed{R_s = \frac{V_i - V_o}{I_z + I_L}}$$

R_s – series resistance

Load Resistance, R_L is given by

$$R_L = \frac{V_o}{I_L}$$

The power P_{zm} (maximum zener power) is given by

$$P_{zm} = I_{zm} \times V_z$$

Comparision of i) zener diode

ii) PN diode

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Zener diode	p-n diode
i) Operated in reverse bias breakdown condition.	i) Never operated in reverse breakdown condition.
ii) Power dissipation capability is high.	ii) low compared to zener diode.
iii) ~~Diode~~ Zener resistance is very small in reverse breakdown condition.	iii) Diode resistance is very high in reverse bias condition.

Comparision of zener breakdown and Avalanche breakdown

Zener ~~diode~~ breakdown	Avalanche breakdown
i) ~~Zener~~ Breaking of covalent bonds is due to intense electric field across the narrow depletion region.	i) Breaking of covalent bonds is due to collision of accelerated charge carriers.
ii) Temperature co-efficient is negative.	ii) Temperature co-efficient is positive.
iii) Breakdown voltage decrease as junction temp increases.	iii) Breakdown voltage increases as temp increases.
iv) This occurs for zener diode when V_z less than 6V.	iv) This occurs for when V_z more than 6V.

Design equation for zener voltage regulator

i) $R_s = \frac{V_o}{I_{L\max}}$ ii) $V_o = V_z$ iii) $I_{z\max} = \frac{P_z}{V_o \text{ (or) } V_z}$

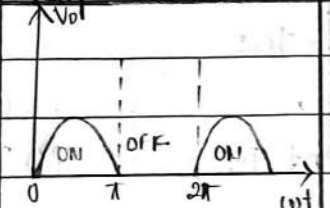
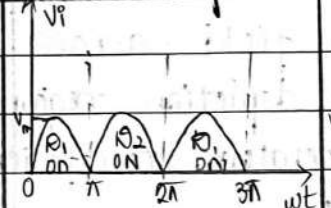
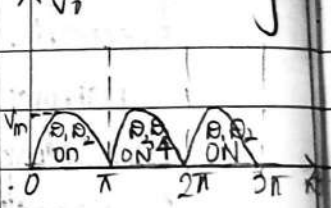
iv) $R_{s\max} = \frac{V_{in\,min} - V_o}{I_{L\max} + I_{z\min}}$ v) $R_{s\min} = \frac{V_{in\,max} - V_o}{I_{L\min} + I_{z\max}}$

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V_{inmin} = minimum i/p voltage
V_{inmax} = maximum i/p voltage
I_{Lmax} = maximum load current
I_{Lmin} = minimum load current
I_{zmin} = minimum zener current
I_{zmax} = maximum zener current.

* If I_{Lmin} is not given choose $I_{Lmin} = I_{Lmax}$
* If I_{min} and I_{max} are not given then choose $I_L = I_{Lmax} = I_{Lmin}$

Comparision of Rectifiers.

	Parameters	Half wave	full wave with C-filters	forward bridge
1.				
2.	I_{av} (or) I_{dc}	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
3.	V_{av} (or) V_{dc}	$\frac{V_m}{\pi}$	$\frac{2V_m}{\pi}$	$\frac{2V_m}{\pi}$
4.	I_{rms} (or) I_{ac}	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
5.	V_{rms} (or) V_{ac}	$\frac{V_m}{\sqrt{2}}$	$\frac{V_m}{\sqrt{2}}$	$\frac{V_m}{\sqrt{2}}$
6.	P_{dc}	$\frac{V_m^2}{R_L \pi^2}$ (or) $\frac{I_m^2 R_L}{\pi^2}$	$\frac{4V_m^2}{R_L \pi^2}$ (or) $\frac{4I_m^2 R_L}{\pi^2}$	$\frac{4V_m^2}{R_L \pi^2}$ (or) $\frac{4I_m^2 R_L}{\pi^2}$

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7.	~~Pac~~ Pac efficiency	$\frac{V_m^2}{4R_L}$	$\frac{V_m^2}{2R_L}$	$\frac{V_m^2}{2R_L}$
8.	efficiency	40% $\eta = \frac{P_{dc}}{P_{ac}}$ (or) $\eta = \frac{V_{ac}^2}{V_{dc}^2 + V_{ac}^2}$	81.03% $\eta = \frac{P_{dc}}{P_{ac}}$	81.03 % $\eta = \frac{P_{dc}}{P_{ac}}$
		R.F (Ripple factor $= \frac{V_{ac}}{V_{dc}}$		
9.	Ripple factor (γ)	1.219	0.4837	0.4837
10.	Peak inverse voltage (PIV)	V_m	$2V_m$	V_m
11.	Transformer	No center tapped (CT) is required	CT required	No CT required
12.	Diode used	1	2	4
13.	Ripple factor for C-filters	$\gamma = \frac{1}{2\sqrt{3} f_c R_L}$	$\gamma = \frac{1}{4\sqrt{3} f_c R_L}$	$\gamma = \frac{1}{4\sqrt{3} f_c R_L}$
14.	Ripple frequency	$f = 50Hz$	$f = 100Hz$	$f = 100Hz$

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